

# Angel Fernando Bórquez Guerrero

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Hermosillo, Sonora.

## Education

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### Universidad de Sonora

Bachelor of Science in Computer Science

GPA: 91.91%

**August 2023 – Present**

### TE Connectivity

AI Engineer Intern

AI Engineering

**June 2026 – September 2026**

- Installed and configured a shared development server for interns, establishing the infrastructure and user access needed to provide a centralized environment for developing and hosting their projects.
- Migrated the orchestration workflow of an AI agent to LangGraph, restructuring its state management and execution flow to improve maintainability and efficiency.
- Reworked the agent's internal architecture and execution flow, reducing response times from up to three minutes to near-instant responses and improving the system's usability.
- Collaborated with project stakeholders to adapt and improve an AI-based system for continued use, translating technical requirements into maintainable software components and workflows.

### ACARUS — Supercomputing Area

Professional Intern

Supervisor: Dr. María del Carmen Heras Sánchez

**September 2025 – Present**

- Professional internship in the Supercomputing area at the University of Sonora.
- Part of a development team building a specialized forum for users of *Yuca*, the university HPC system.
- Implementing features such as user posts, direct messaging, and service request workflows.
- Supporting an active HPC user community by improving communication and technical support processes.

## Research Experience

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### University of Guadalajara

Research Intern

Supervisor: Dr. Rafael Morales Gamboa

**June – July 2025**

- Generalized and validated an Entropic Associative Memory (EAM) model across multiple datasets through large-scale experimental evaluation and robustness analysis.
- Adapted and refactored the data pipeline to support class balancing, dynamic class selection, and scalable dataset handling.
- Adapted perceptual neural networks (autoencoder and classifier) to generate feature representations compatible with the EAM architecture.
- Designed and implemented a novelty detection experiment to evaluate the model's ability to reject unseen classes.
- Introduced experimental controls enabling partial memory registration and quantitative analysis of rejection behavior via no-response metrics.
- Refactored the system into a flexible experimental framework allowing controlled evaluations with a variable number of classes.

## Competitions & Projects

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### **NASA Space Apps Challenge 2024**

Project Co-Lead

Project: Voyager CXXV

**October 2024**

- Co-led the development of a geospatial simulation tool to evaluate the impact of urban green areas on temperature and air quality in Hermosillo, Mexico.
- Designed and implemented a system allowing users to define custom zones and simulate long-term environmental effects of urban reforestation.
- Integrated spatial data visualization through heat maps to support data-driven insights for urban planning scenarios.
- Coordinated technical tasks within the team, contributing to project planning, feature definition, and final integration.

### **TE AI CUP 2025 – International 3rd Place**

UI Developer

Project: AI Generated Machine Downtime Insights

**January – May 2025**

- Designed and developed the UI/UX for an AI-based industrial downtime analysis platform, translating complex machine time-series data and unsupervised learning outputs into clear, actionable visual insights.
- Developed an accessible and user-centered interface for non-technical users, simplifying the exploration and interpretation of AI-generated insights without requiring prior knowledge of machine learning.
- Defined user flows, information hierarchy, and interaction patterns to improve navigation and facilitate the transition from high-level downtime trends to detailed analytical results.
- Collaborated with the data science and engineering teams to align the interface with the underlying analytical models and ensure that visual representations accurately reflected model outputs.

### **NASA Space Apps Challenge 2025**

Project Co-Lead

Project: Endurance CXXV

**October 2025**

- Co-led the development of an educational weather application designed for a children-oriented audience, focused on climate awareness and engagement.
- Developed both frontend and backend components of the application, contributing to core functionality and system integration.
- Designed and implemented the user interface with an emphasis on accessibility, usability, and age-appropriate interaction.

### **MICAI 2025 MeDA Challenge – National 1st Place**

**October 2025**

- Contributed to a 1st Place victory in a national academic competition focused on Medical Domain Adaptation using Self-Supervised Learning (SSL), leading to an invitation to publish in a special issue of the *Health Information Science and Systems* journal.
- Structured and standardized the dataset and training assets to establish a reproducible data pipeline for efficient model development and experimentation.
- Investigated approaches for cross-domain generalization and representation learning across heterogeneous medical imaging datasets.
- Developed visual and presentation assets used to communicate the project during the competition and dissemination stages.

### **SmartCity Hackathon – People’s Choice Award**

Frontend Developer

Project: SmartTransit

**March 2026**

- Developed the main frontend interface for an AI-based public transportation optimization system using demographic, geographic, and urban growth variables.
- Designed the UI structure and internal frontend logic to support route generation and optimization workflows.

## **HSIL Hackathon 2026 – 3rd Place**

Application Designer & AI Agent Architect

Project: ShiftGuardian

**April 2026**

- Designed the application interface and AI agent architecture for a clinical decision support system supporting nurse handoffs and patient prioritization.
- Integrated a RAG-based LLM workflow concept to summarize clinical reports, identify critical tasks, and reduce cognitive load during shift transitions.

## **TE AI CUP 2026 – International 1st Place**

Project Leader

Project: Industrial Alert Agent

**January 2026 - June 2026**

- Led a 6-member interdisciplinary team in an international AI competition, guiding the technical development of an AI-driven system for industrial alert analysis that achieved 1st place.
- Developed an AI agent integrating Large Language Models (LLMs) for automated alert analysis and decision support.
- Developed an end-to-end data and machine learning pipeline involving data cleaning, text preprocessing, embedding generation, clustering, data validation, and predictive modeling.
- Developed a predictive modeling system to estimate the likelihood of future industrial alerts, providing a proactive component that complemented the analysis of historical patterns and enabled earlier identification of potential issues.
- Architected the project around clean, standardized, and scalable software structures, establishing conventions for modules, tools, configuration, state management, and component integration.

## **Technical Skills**

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**Languages:** Python, C++, Java, SQL, JavaScript, TypeScript

**Frontend:** HTML, CSS, SASS, Vue.js, NPM

**Backend & Frameworks:** Django

**Systems & Infrastructure:** Linux (Arch), HPC, Virtual Machines, Proxmox VE, VirtualBox

**Tools:** Git, GitHub, Jupyter, Google Colab, LaTeX